IBM Innovation Centre for Education

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Abstract :
The Anomaly Detection For Healthcare Transaction

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Introduction:

Healthcare organizations process a large number of financial transactions every day, including insurance claims, billing records, reimbursements, and provider payments. Monitoring these transactions manually is difficult because of the large volume of data. This may result in delayed fraud detection, billing errors, and financial losses. To overcome these challenges, an automated anomaly detection system is required.
The Healthcare Transaction Anomaly Detection System is designed to identify suspicious or unusual transactions automatically by analyzing healthcare datasets. The system helps organizations quickly detect abnormal claim amounts and irregular billing activities through a simple and efficient web-based application.

Problem Statement

Manual checking of healthcare transactions is time-consuming and less accurate. Fraudulent claims, duplicate payments, and abnormal transaction amounts may go unnoticed. Therefore, there is a need for a smart system that can automatically analyze healthcare transaction data and identify anomalies.

Objective of the Project

- To detect suspicious healthcare transactions automatically.
- To reduce manual effort in transaction verification.
- To improve accuracy in fraud detection.
- To provide an easy-to-use interface for data upload and analysis.
- To generate quick anomaly detection results.

Proposed System

The proposed system uses Python-based technologies to analyze healthcare transaction data. Users can upload a CSV file containing transaction details. The system reads the dataset, processes the data, and applies anomaly detection techniques such as the Z-score method. Transactions that significantly differ from normal patterns are marked as suspicious.

Modules of the Project:

1. User Interface Module

Provides a web-based interface using Streamlit where users can upload files and view results.

2. Data Upload Module

Accepts healthcare transaction data in CSV format and loads it into the system.

3. Data Preprocessing Module

Checks for missing values, cleans data, and prepares it for analysis.

4. Anomaly Detection Module

Uses statistical calculations to identify unusual transaction records.

5. Result Module

Displays suspicious transactions and processed output in table format.

Technologies Used

- Python
- Streamlit
- Pandas
- NumPy
- Jupyter Notebook
- VS Code

Folder Enhancement Structure:

```
Healthcare_Transaction_Anomaly_Detection/
│ ── app.py
│ ── requirements.txt
│ ── README.md
│
├── dataset/
│   └── healthcare_transactions.csv
│
├── notebooks/
│   └── test.ipynb
│
├── src/
│   ├── preprocessing.py
│   ├── detection.py
│   └── utils.py
│
├── outputs/
│   ├── anomaly_results.csv
│   └── reports/
│
├── images/
│   └── screenshots/
│
└── docs/
    └── synopsis.docx
```

Input and Output:

Input:

Healthcare transaction dataset in CSV format containing fields such as patient ID, provider ID, claim amount, claim date, and transaction details.

Output:

- Full processed dataset
- Suspicious transaction records
- Anomaly detection report
- Transaction classification results

Advantages:

- Fast processing of large datasets
- Reduces manual work
- Improves fraud detection efficiency
- Easy to use system
- Cost-effective solution

Limitations:

- Works mainly on numerical anomalies
- Requires clean and structured data
- Complex fraud patterns may need advanced machine learning models

Future Enhancements:

- Integrate Machine Learning algorithms (Isolation Forest, Autoencoders)
- Real-time fraud monitoring dashboard
- Email alerts for suspicious transactions
- Database integration
- Advanced visual analytics

Conclusion:

The **Healthcare Transaction Anomaly Detection System** is an efficient solution for identifying suspicious healthcare transactions. By using statistical anomaly detection techniques and a user-friendly interface, the system improves auditing efficiency, reduces fraud risks, and supports better financial management in healthcare organizations.